## 1.1.a. The structure and function of the skeletal system

Learners will be able to name and locate the major bones of the body and be able to apply examples of how the skeletal system allows the functions such as posture and protection.

Learners will be able to identify major joints along with the associated articulating bones in the knee, elbow, shoulder and hip. Knowledge will be developed of the types of movement at hinge joints and ball and socket joints, as well as being able to apply these movements to examples from physical activities and sports.

| Topic area | Learners must: |
|---|---|
| Location of major bones | • know the name and location of the following bones in the human body:<br>○ cranium<br>○ vertebrae<br>○ ribs<br>○ sternum<br>○ clavicle<br>○ scapula<br>○ pelvis<br>○ humerus<br>○ ulna<br>○ radius<br>○ carpals<br>○ metacarpals<br>○ phalanges<br>○ femur<br>○ patella<br>○ tibia<br>○ fibula<br>○ tarsals<br>○ metatarsals. |
| Functions of the skeleton<br> | • understand and be able to apply examples of how the skeleton provides or allows:<br>○ support<br>○ posture<br>○ protection<br>○ movement<br>○ blood cell production<br>○ storage of minerals. |
| Types of synovial joint | • know the definition of a synovial joint<br>• know the following hinge joints:<br>○ knee – articulating bones – femur, tibia<br>○ elbow – articulating bones – humerus, radius, ulna<br>• know the following ball and socket joints:<br>○ shoulder – articulating bones – humerus, scapula<br>○ hip – articulating bones – pelvis, femur. |

| **Topic area** | **Learners must:** |
|---|---|
| Types of movement at hinge joints and ball and socket joints<br> | • know the types of movement at hinge joints and be able to apply them to examples from physical activity/sport:<br>○ flexion<br>○ extension<br>• know the types of movement at ball and socket joints and be able to apply them to examples from physical activity/sport:<br>○ flexion<br>○ extension<br>○ rotation<br>○ abduction<br>○ adduction<br>○ circumduction. |
| Other components of joints | • know the roles of:<br>○ ligament<br>○ cartilage<br>○ tendons. |